



Beyond the Bite

What shark incident data really tells us

Team Serie: Ailixier, Amanda, Berker & Tina

Challenging the myth



We are **Save Our Sharks**, a U.S. shark conservation NGO running an **awareness campaign** to challenge the perception that sharks are bloodthirsty killers.

- **Public fear of sharks** is greater than the actual danger they present. Since the release of *Jaws* in 1975, sharks have been widely feared and misunderstood. In reality, only a small fraction of shark species pose any risk to humans, while sharks play a vital role in maintaining healthy ocean ecosystems.
- Using the **Global Shark Attack dataset**, we will investigate the fatality rate and understand if these shark attacks have been provoked or unprovoked by humans. We will compare the shark incidents over the last 50 years to see if there has been an increase. We aim to understand if this shark aggression is caused by growing human populations, greater beach activity, and increased overlap between humans and shark habitats.

Our goal is to separate fear from fact and better understand the true nature of shark-human interactions.

Understanding the data



- We looked at the [Global incident log of shark attacks](#) dataset of 7090 entries and discussed which columns to include and exclude in our analysis.
- We identified the following columns based on the questions we wanted to answer:
 - Type - What proportion of shark incidents are unprovoked vs. provoked?
 - Fatal - How many incidents are actually fatal?
 - Species - Which species are involved — and are they truly dangerous?
 - Species - Which species appear in fatal vs. non-fatal cases?
 - Year - Has the number of incidents increased over time — and if so, why?
 - Gender - Does gender play a role in shark incident patterns?
 - Activity - What are the types of the incidents?
 - State - Is there a state with a much larger percentage of incidents?

Dataset problems



- On closer inspection we realised that there were some problems with the dataset, namely:
 - Missing values
 - Understanding how to deal with the missing values in df.
 - Formatting issues
 - Understanding if the data is standardized and readable.
 - Inconsistent data
 - Understanding if the data is accurate.

Cleaning process

- We identified the main columns we needed for analysis and applied the following **cleaning techniques**:
 - Removed white space
 - Replaced missing values with Unknown
 - Combined similar values together
 - Standardized everything to lowercase
 - Dropped columns we didn't need
- We made some important **transformations** for example:
 - Converted results into percentages to be able to better analyse the data
 - Grouped Species into 4 main categories
 - Using pivot tables to get a better view of the results
- **Structural** decisions
 - Filtering out only USA, excluding territories (for example Puerto Rico, virgin islands)
 - Grouping the years to before and after 1975

Data structuring



- We created new variables in our data frame in order to support our analysis
- Changed data type where necessary, for example, from object to string
- Groupings and aggregations techniques to organize the data around the key questions we wanted to answer
- Final dataset organization: Clean, structured, and ready for analysis

Major challenge



- Qualitative data in open text fields makes analysing and reporting on data very difficult because of the large number of unique entries, for example “injury” and “species”.
- Some of the values under "Type" column were not clear - provoked, unprovoked. So, we looked at "injury" and "species" column and there were some useful information which lead us to drop some values completely from the data frame.
- Country and State had messy city/country names.
- Analysing data that has a correlation does not necessarily mean causation. For example, more incidents happened after the Jaws was released, does not mean that the movie is responsible for this. Understanding that we don't know the answers to all the Why's and that often further research is needed.

Final insights



Total Shark Attacks in USA	Fatality Rate	Survival Rate	Most Fatal Activity	Most Fatal State	Most Lethal Species
36.67%	7.35%	92.65%	Swimming 16.28%	Florida 46.92%	Tiger Shark 10.88%

Fatality Breakdown

	Metric	Value
Gender	Male Fatality Rate	8.36%
	Female Fatality Rate	3.74%
Attack Type	Provoked Fatality Rate	1.29%
	Unprovoked Fatality Rate	8.45%

Potential applications

- Launch the awareness campaign in Florida, since they have the highest percentage of shark incidents
- Conduct further research on shark migration patterns for the bull shark, tiger shark and white shark to better understand these species that are involved in many incidents
- Use verified statistics to debunk the myth of "increasingly aggressive" shark behavior.

Conclusions

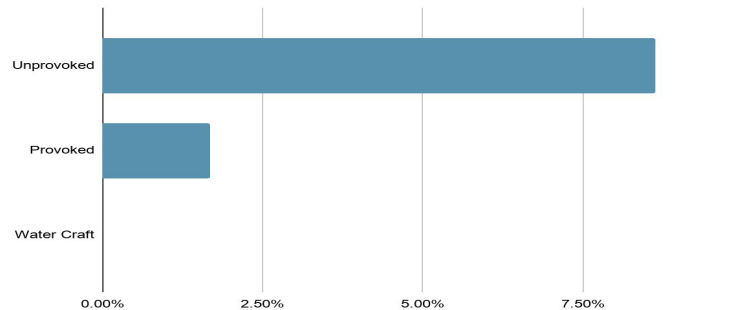
Our data shows that most of the encounters are **not fatal**. Unprovoked incidents carry a fatality rate of 8.45%, Although, it is 6 times higher than provoked incidents at 1.29%, it's only fatal at this level. If sharks were bloodthirsty killers, they wouldn't leave almost 92% of their unprovoked encounters alive. Even the most feared encounters are rarely fatal.

Final insights



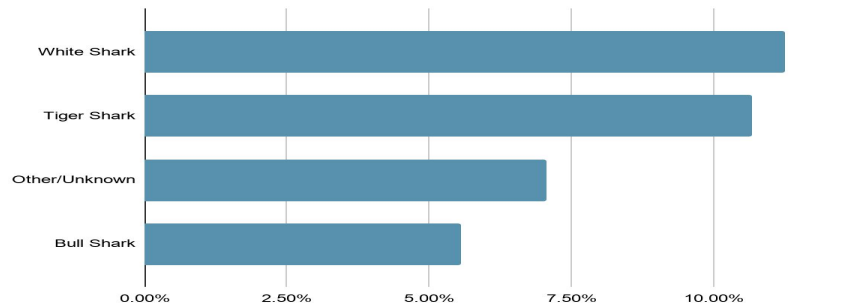
Type: What proportion of shark incidents are unprovoked vs. provoked? And what is the fatality rate?

fatal y/n	n	y	Total	Fatality Rate (%)
type				
Watercraft	80	0	80	0.00
Provoked	229	3	232	1.29
Unprovoked	1799	166	1965	8.45



Species: Which species are involved — and are they truly dangerous?

fatal y/n	n	y	Total	Fatality Rate (%)
species				
Bull Shark	83	5	88	5.68
Tiger Shark	131	16	147	10.88
White Shark	205	25	230	10.87
Other / Unknown	1689	123	1812	6.79



Final insights



Gender - Does gender play a role in shark incident patterns?

Fatality rate by gender:

fatal y/n	n	y	Total	Fatality Rate (%)
sex				
F	360	14	374	3.74
M	1656	151	1807	8.36

Fatality rate by gender:

fatal y/n	n	y	Total	Fatality Rate (%)
sex				
F	360	14	374	3.74
M	1656	151	1807	8.36

Year - Has the number of incidents increased over time — and if so, why?

Number of incidents BEFORE 1975: 334

Number of incidents AFTER 1975: 1727

Final insights



Activity - What are the types of the incidents? Surfing and swimming

Activity breakdown:

	Incidents	Percentage (%)
activity_group		
Surfing	784	34.3
Swimming	396	17.3
Fishing	308	13.5
Other	225	9.9
Wading / Standing	200	8.8
Diving / Snorkeling	139	6.1
Unknown	108	4.7
Bodyboarding	70	3.1
Boating/Paddling	53	2.3

Activity vs Fatality Rate:

fatal y/n	n	y	Total	Fatality Rate (%)
activity_group				
Surfing	765	14	779	1.80
Swimming	329	64	393	16.28
Fishing	283	25	308	8.12
Other	190	32	222	14.41
Wading / Standing	195	4	199	2.01
Diving / Snorkeling	124	14	138	10.14
Unknown	94	5	99	5.05
Bodyboarding	64	6	70	8.57
Boating/Paddling	51	2	53	3.77

Changing the narrative



Before the data:

"Sharks are bloodthirsty killers."

After the data:

"Sharks are wild animals sharing an ocean with an increasing number of humans."

By replacing fear with facts, we can help build public support for shark conservation and protect a species that plays a vital role in marine ecosystems.

Save Our Sharks

Protecting sharks starts with understanding them.

Thank You 🐟

Ailixier, Amanda, Berker & Tina